TECMILENIO

# ACTIVITY 3


**Nombre:**

- David Moreno
- Gustavo Martinez
- Emilio Gallegos
- Diego Moya

**Matrícula:**

- AL03081455
- AL02962078
- AL03107568
- AL03089635


**Materia:** Knowledge Management Systems and Technologies

**Maestro:** Alfredo Paul Aguirre Torres

**Tarea:** Activity 3

**Fecha:** 06/11/2026

*Top-K Document Recommender with Cosine Similarity — Nexo IT Services*

**Activity 3**

# Document recommendations

Nexo IT Services

## Company and objective

Nexo IT Services is a fictional support provider whose analysts repeatedly search for access, backup and change procedures. I used 30 simulated English documents to test whether the system could find useful material for a support analyst. Each CSV row has a unique id, title and text, plus category, topic, concepts, owner, version and review date. The repository covers Policy/Process, Support/Incidents and Knowledge/Training.

## Method: TF-IDF

The Python script normalizes case and accents, replaces special characters, removes a documented English stopword list, and represents unigrams and bigrams with TfidfVectorizer. Sublinear term frequency reduces the effect of repeated words, smooth inverse document frequency weights terms, and L2 normalization enables cosine comparison. The fitted repository contains 2,317 features. These are lexical vectors rather than semantic embeddings.

An ID supplies its document vector and excludes that document from the results. Free text uses the same fitted vocabulary. Results sort by descending cosine similarity with a deterministic ID tie-break. Both modes return five documents. The script validates empty queries, unknown IDs, missing files, missing columns and duplicate identifiers. An out-of-vocabulary query yields zero scores and should be rephrased.

## Test results

**1. Policy/Process seed** - Input: DOC002

| Rank | Document | Cosine |
|---|---|---|
| 1 | DOC004 - Remote connection troubleshooting FAQ | 0.1929 |
| 2 | DOC003 - VPN failure following credential rotation | 0.1799 |
| 3 | DOC001 - Remote access policy | 0.1693 |
| 4 | DOC026 - Stale authentication token recovery | 0.0965 |
| 5 | DOC005 - Secure remote working workshop | 0.0725 |

**2. Support/Incidents seed** - Input: DOC008

| Rank | Document | Cosine |
|---|---|---|
| 1 | DOC007 - File restoration operating procedure | 0.0930 |
| 2 | DOC009 - Deleted files recovery FAQ | 0.0813 |
| 3 | DOC006 - Backup retention policy | 0.0747 |
| 4 | DOC010 - Quarterly restoration exercise | 0.0617 |
| 5 | DOC029 - Monitoring alert caused duplicate tickets | 0.0465 |

**3. Free-text query** - Input: restore deleted customer files after a failed backup

| Rank | Document | Cosine |
|---|---|---|
| 1 | DOC007 - File restoration operating procedure | 0.3216 |
| 2 | DOC009 - Deleted files recovery FAQ | 0.2188 |
| 3 | DOC010 - Quarterly restoration exercise | 0.1900 |
| 4 | DOC006 - Backup retention policy | 0.1259 |
| 5 | DOC008 - Backup job failed because storage was full | 0.0966 |

## Discussion of the results

**Test 1.** The seed concerns VPN recovery. Related access policy, incident and FAQ share the same recovery task. Training and onboarding are broader lexical neighbors and require a relevance check.

**Test 2.** A failed backup should retrieve retention and restoration material. A shared term alone does not guarantee an actionable recovery instruction.

**Test 3.** The query joins deletion, customer files and a failed backup. Inspect the restore procedure and FAQ before applying a recovery point.

## Use in the support team

The results bring an incident record together with the policy or procedure that explains how to handle it. For example, the backup query ranks DOC007 first (0.3216), followed by recovery guidance and a training exercise. An analyst could use these links to find a restoration procedure more easily. I did not measure search time or ticket resolution, so this remains an expected benefit.

## Limitations and possible improvements

**Synonyms and ambiguity.** DOC026 uses "corporate tunnel", "secret" and "stale token" rather than the VPN query vocabulary. A lexical search can miss operationally relevant guidance. DOC005 contains words from the query even though its training content supplies no corrective procedure.

**Document length and corpus bias.** Long articles can dilute a specific instruction. Thirty deliberately structured documents are too small and homogeneous to represent an operational service desk. Scores indicate similarity rather than correctness or authorization.

**Next iteration.** Add controlled synonyms and compare embeddings on a separate query set. Split long documents into traceable sections, rank approved current procedures above generic training, and filter by service, owner and role on the server. Evaluate multiple query intents with independently reviewed relevance labels. Track false positives, missed procedures and outdated articles continuously.

## Code and evidence

Run **python recommender.py --id DOC002** or **python recommender.py --query "restore deleted customer files after a failed backup"**. The README lists dependencies and setup. The evidencias/activity3 folder contains the three raw execution logs and three browser screenshots. The same corpus and unmodified retriever continue into Activity 4.

## Reflection video

Open the narrated reflection video. The video uses synthetic English narration.

Prototype and resources: Nexo IT Knowledge Lab.

## Technical references

scikit-learn. TfidfVectorizer. Official API documentation.